

Food Standards Australia New Zealand (FSANZ) guidelines for raw milk collection advise milk should be cooled on farm to 5°C within 3.5 hours from the start of milking. FSANZ guidelines also allow manufacturers to collect milk at times and temperatures other than 5°C after 3.5 hours provided the manufacturer undertakes a risk assessment of the milk and processes it accordingly to ensure its safety.

## Validation of food safety for early collection of raw milk

Milk may be picked up from farms earlier at a higher temperature provided you are able to manage the food safety risks.

The scientific modelling work supports that milk can be picked up earlier, at a higher temperature, as long as it is on or below the time-temperature curve in either of these two diagrams.

Where milk is collected at time-temperature that does not meet the FSANZ guidelines or does not meet the milk cooling curves, a risk assessment to ensure the safety of the milk must be conducted. Tables 1 and 2 provide a validated example of how to do this risk assessment.

Evidence you must have to support the risk assessment includes:

- time after milking commenced
- temperature of milk at collection
- initial bacterial load (e.g. *B. cereus*) in the vat at milk collection or information that can be used to determine/correlate to the growth of bacteria of concern.

## Milk cooling curve

The University of Tasmania undertook scientific modelling work for the Australian dairy industry in 2014 to assess the food safety of milk collected at times and temperatures outside the FSANZ guidelines.

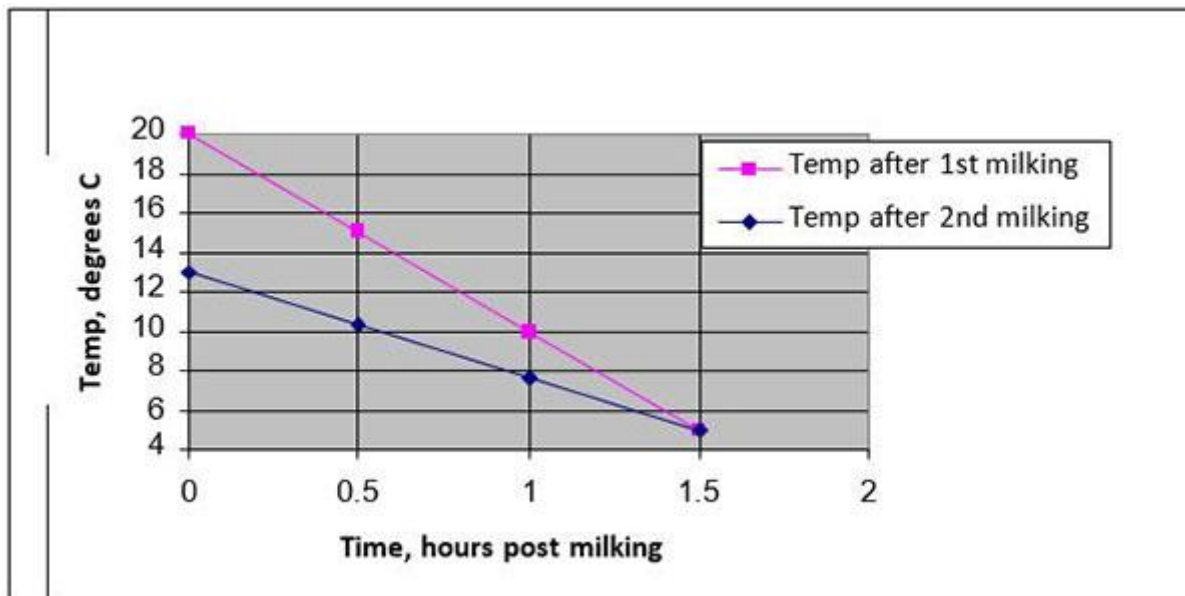
When raw milk is collected from a dairy farm before it is cooled to 5°C within 3.5 hours, or 8°C within 2 hours for the EU, diagrams 1 and 2 can be used to assess whether the milk is acceptable within food safety parameters. Provided the milk temperature and time of pick up falls below the curves, the milk cooling curve determines the product as safe and no further assessment is needed.

If the milk temperature doesn't fall within these parameters, your establishment must separately validate the safety of milk and have records to support the decision.

## Food Standards Australia New Zealand (FSANZ) guidelines

The cooling envelope assumes the average time to milk a herd is 2 hours and the maximum time available for cooling after milking is therefore 1.5 hours to meet the FSANZ guidelines for milk to be cooled to 5°C by 3.5 hours from the commencement of milking.

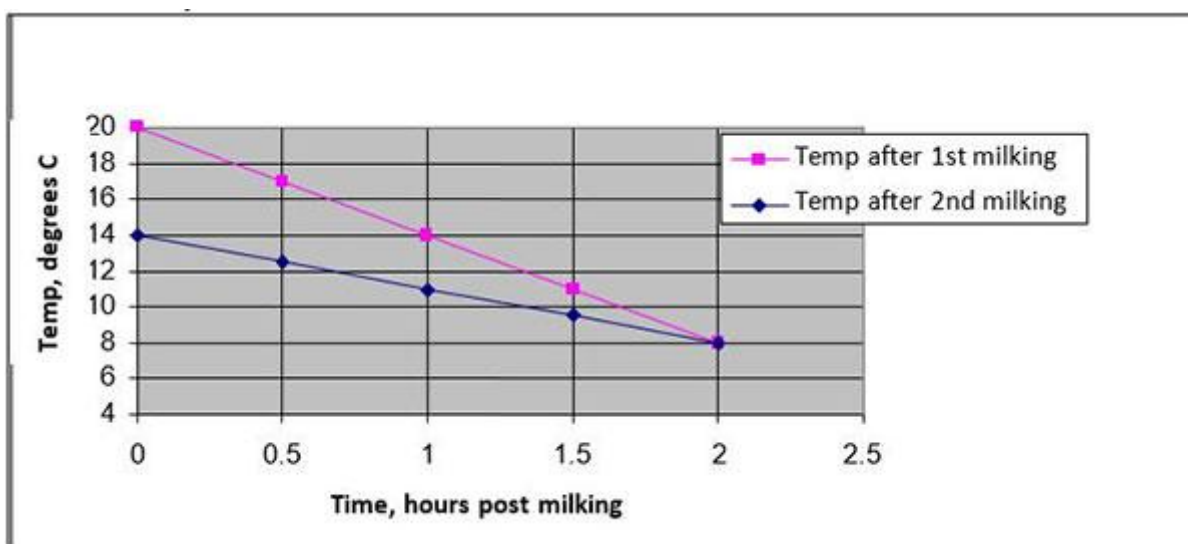
**Diagram 1: Post milking cooling envelope to meet FSANZ guidelines for milk**



## European Community (EC)/Codex for heat treated milk products

The EC/Codex requirement is milk shall be cooled to 8°C within 2 hours after milking. Diagram 2 is based on an assumed maximum temperature of vat milk of 20°C at the end of the first milking and 14°C at end of the second milking.

**Diagram 2: Post-milking cooling envelope to meet EC/Codex requirements for heat treated milk products**



## Assessing the safety of milk that doesn't fit the curve

Milk collected at temperatures higher than regulatory requirements or which sits outside the milk cooling envelopes may be safe as bacterial load may not have increased to a point where pathogen levels

or spoilage organisms will cause illness in humans (10<sup>5</sup> cells/ mL for *B. cereus* or 10<sup>7</sup> cells/ mL for *Pseudomonas*).

Note: The data in tables 1 and 2 assume milk is held at the same temperature from the time of pick-up.

Milk processors/dairy manufacturers can compare their information to these tables to decide from which farm milk can be picked up early or on a case by case basis. The safety of the milk can be assessed to determine the bacterial growth over a period of time at a given starting temperature.

To be able to do a risk assessment about the potential growth of pathogens and spoilage organisms, you must have data on the level of bacteria at time of milk collection, the time milk has been held since milking and milk pickup temperature.

**Table 1: Estimated time (hours) before the threshold level is reached for *B. cereus* based on initial concentration (cells/ mL) measured as a logarithmic phase**

| Temperature (°C) at time of collection | 1 log  | 2 log  | 3 log  | 4 log |
|----------------------------------------|--------|--------|--------|-------|
| 20                                     | 15.5   | 11.6   | 7.7    | 3.9   |
| 15                                     | 29.1n  | 21.8   | 14.5   | 7.3   |
| 10                                     | 73.3n  | 55.0n  | 36.6n  | 18.3  |
| 8                                      | 125.3n | 94.0n  | 62.7n  | 31.3n |
| 6                                      | 261.5n | 196.1n | 130.7n | 65.4n |

n time (hours) before milk at the specified temperature would reach the concentrations of (10<sup>5</sup> cells/ mL for *B. cereus*), assuming an initial concentration in the vat at the time of milk pick-up of 1 log *B. cereus* in the first column increasing to 4 log in the final column.

**Table 2: Estimated time (hours) before the threshold level is reached for *Pseudomonas*, based on initial concentration (cells/ mL) measured as a logarithmic phase**

| Temperature (°C) at time of collection | 1 log | 2 log | 3 log | 4 log |
|----------------------------------------|-------|-------|-------|-------|
| 20                                     | 18.6  | 14.9  | 11.2  | 7.5   |

|    |       |       |       |       |
|----|-------|-------|-------|-------|
| 15 | 27.3n | 22.2  | 17.0  | 11.9  |
| 10 | 44.8n | 36.8n | 28.7n | 20.7  |
| 8  | 57.0n | 46.9n | 36.8n | 26.7n |
| 6  | 74.9n | 61.9n | 48.8n | 35.7n |

n time (hours) before milk at the specified temperature would reach the concentrations of concern (10<sup>7</sup> cells/ mL for *Pseudomonas*), assuming an initial concentration in the vat at the time of milk pick-up of 1 log *Pseudomonas* in the first column increasing to 4 log in the final column.

## Non-compliant farm milk

Corrective action must be taken in accordance with your documented procedures. Advise the farm of the unsatisfactory result and ensure action is taken to correct the problem. You may decide ongoing failures by the farm will result in the milk not being collected. If so, a policy of this nature needs to be fully detailed in your approved arrangement (AA).

Your procedures must also ensure corrective action is demonstrated when high temperatures are identified and steps are taken to verify the corrective action has been effective. These procedures must be documented in your AA will be subject to audit.

## Ineligible for export

If the milk or ingredients do not comply with raw milk collection time and temperatures, the milk cooling curve or further validation then they must be excluded from export and your declaration of compliance and transfer declarations must indicate the goods are not export eligible.

## Demonstrating compliance

The minimum procedures that should be documented in the AA include:

- temperatures under which raw milk will be collected from farms (FSANZ guidelines, importing country or milk cooling curve limits)
- temperatures under which milk will be received at the establishment
- triggers for corrective action when limits are exceeded and what corrective action will include
- how non-compliant milk will be managed
- how export ineligible milk is identified in the system.

Records that must be kept include:

- temperature of raw milk collected at farm and received at the establishment
- final product declarations of compliance and transfer declarations including a statement about specific country eligibility or otherwise.

## **Requirements for dairy ingredients and milk received from another establishment**

Export eligible establishments must ensure their systems demonstrate the dairy ingredients used to produce milk and milk products for export:

- met temperature requirements for the raw milk component stage of production. This includes imported ingredients or those used as part of contract manufacturing arrangements.

When sourcing dairy ingredients or liquid milk from another establishment or overseas:

- obtain evidence from the supplier indicating the milk used in the production of dairy products complies with the specific importing country requirements for temperature.

If you cannot obtain or verify this information the products you produce are not eligible for export.

Source: <https://www.agriculture.gov.au/biosecurity-trade/export/controlled-goods/dairy/registered-establishment/raw-milk-temperatures>